

TREMORTRACE · DESIGN DECISIONS REGISTER

TremorTrace design decisions register

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DOCUMENT CONTROL

Document ID	TRT-DEC-001	Version	0.1
Status	Draft	Date	2026-10-02
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/tremortrace		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-02	Amish Chadha	Register opened; open decisions from the review note, the decision records and the build plan work

TremorTrace design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` or in the review note (`docs/REVIEW.md`); this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

Safety: TremorTrace is a research and educational prototype, not a medical device. It has not been cleared or approved by any regulator and must not be used to diagnose, treat or monitor any person.

■ Open decisions

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Accept the design-for-construction changes C1 to C9 (spring bar tip holes, four corner screws, deeper notch, glue collar, cell ribs, foam pads, cell plug removed, light pipe, 0.8 mm gasket)	Accept; accept some; revert to the concept	Accept: each keeps what the band does and makes a part buildable	The whole build plan follows them	TRT-DDR-003, Table 1
2	Firmware update route once the pod is sealed	(a) over Bluetooth with the nRF52840's over-the-air bootloader; (b) open the pod for each update	(a), keeping (b) for recovery	Whether firmware must be loaded before step 8 of the build plan only once, or every time	TRT-DDR-003, A1
3	Protection for the skin-side charging contacts	(a) confirm the module's 5 V pin is dead with no charger and add nothing; (b) add a Schottky diode in the positive lead in any case	(a), with (b) if the pin is found live	Wiring step 3 (one more part in one lead)	TRT-DDR-003, A2
4	First co-design and validation partner	A movement disorders clinic, a patient association, or the OpenRatio network	None made	Not part of the TRL 3 build	TRT-DDR-001, O1; TRT-DDR-002
5	Confirmation of OpenRatio as a possible partner	Confirm; do not name	None made	None	TRT-DDR-001, O2
6	Approval of kit 1.3.1, the forearm-for-scale rule for small objects in concept media	Approve; keep the 1.75 m person	None stated	Concept media only	TRT-DDR-001, O3
7	Strap path in the appearance model	Keep the circular strap of <code>model.py</code> for the drawing and mass; follow the elliptical clay wrist only in renders	Keep the circle in <code>model.py</code> ; elliptical path for renders only	None	<code>docs/REVIEW.md</code> , 2026-09-26, item 1
8	Lid marking and grip ribs shown in the renders but not in the model or BOM	Keep as printed-in features; drop	Keep, with the mark raised or debossed 0.14 mm; no cost change	Lid and base prints, if adopted	<code>docs/REVIEW.md</code> , 2026-09-26, item 3
9	Module detail and charging lead in the renders (not taken from a datasheet)	Accept as illustration only; model from datasheets	Accept as illustration only	None	<code>docs/REVIEW.md</code> , 2026-09-26, item 4

#	Decision needed	Options	Recommendation	Affects in the build	Source
10	How the analysis separates tremor during a reach from voluntary movement	Keep the separate activity flag and report both (current); a better classifier later	Keep the current record and decide the classifier with recorded data at TRL 4	Firmware and notebook only	TRT-CAL-001, section 4; TRT-PRC-001

Item 2 of the 2026-09-26 appearance review (pan-head screws standing above the lid) is answered by change C2 of TRT-DDR-003 (countersunk screws, flush), which is itself open as item 1.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The status LED position on the module	It sets the light pipe bore (6.5 mm from the centre on the centre line) and the hole in the upper foam pad	TRT-DDR-003, C6 and C8
2	The module's thickness with its components (3.5 mm assumed)	It sets the squeeze on the upper foam pad; more than 4.5 mm would need a thinner pad	TRL 3 model; TRT-CAL-001, section 7
3	The charging receptacle's body size (4 x 8 x 3 mm assumed) and whether it has a flange	It sets the floor opening and the glue collar	TRT-DDR-003, C4
4	The spring bars' tip diameter (about 0.9 mm)	The horn holes are drilled 1.0 mm; tips over 1.0 mm need a larger drill	TRT-DDR-003, C1
5	That the module's 5 V pin carries no voltage with the cell connected and no charger	The receptacle contacts touch the skin; see open decision 3	TRT-DDR-003, A2
6	That a printed horn takes an M2 thread-forming screw without splitting (1.2 mm of plastic round the screw)	If it splits, print the horns hotter or use M1.6 screws	TRT-DDR-003, C2
7	The textile strap's mass (8 g assumed) and that its ends loop round a 1.5 mm bar inside a 4.5 mm notch	Mass margin under R6 is 3.8 g; the loop clearance is 0.35 mm	TRT-CAL-001, section 7; TRT-DDR-003, C3
8	The IMU sample-rate tolerance and the module's run current	They carry the R1 and R4 results	TRT-CAL-001, sections 2 and 5

■ Value engineering

Value-engineering target: USD 150 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 43.69 (USD 106.31 under the target). Main cost drivers and savings worth trying:

- The largest lines are the controller and IMU module (USD 15.99), the textile strap (USD 10.00), the magnetic charging receptacle with its cable (USD 6.00) and the LiPo cell (USD 5.95); together they are USD 37.94 of the USD 43.69.
- Making the design constructable added USD 0.50 (two more lid screws); the foam pads and epoxy fall within the existing consumables allowance.

- Savings worth trying: a generic 22 mm textile strap (often under USD 5); buying the charging receptacle and cable as a bulk pair; the module price in small quantities from a distributor.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	D1 single nRF52840 module with built-in IMU, no custom PCB; D2 10 s summaries with optional raw snippets; D3 web Bluetooth page first; D4 most affected wrist first; D5 local-first data	Amish: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them."	TRT-DDR-001
2026-09-25	D6 charge at the 50 mA setting; D7 woven textile strap of about 8 g	Amish: "i accept all your recommendations, go with them across all repos."	TRT-DDR-002
2026-09-30	Make the design physically buildable while drawing the build plan; keep open decisions out of the build plan	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations." The resulting changes are open as item 1	TRT-DDR-003
2026-10-01	Treat <code>budget_usd</code> as a value-engineering target, not a limit	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens"	<code>.kit/STANDARDS.md</code> , section 18